Statistics and Data Science 188 / *Statistiek en Datawetenskap 188*

Chapter 3 calculator instructions / *Hoofstuk 3 sakrekenaar instruksies*

1. Changing number of decimals. / *Verander aantal desimale*.
2. Clearing the calculator memory. / *Maak sakrekenaar geheue skoon*.

3. Reading in data (one variable). / *Inlees van data (een veranderlike)*.

X:	-6.1	-2.8	4.3	7.6	12.9

4. Obtain / *Verkry*

- Sample size / *Steekproefgrootte* (*n*=5)
- Summations / *Sommasies* $\sum_{i=1}^{n} x_i = 15.9$, $\sum_{i=1}^{n} x_i^2 = 287.71$
- Average / *Gemiddelde* $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 3.18$
- Standard deviation / *Standaardafwyking* $s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = 7.6998$

5. Reading in data (two variables). / *Inlees van data (twee veranderlikes)*.

X:	1.2	1.8	1.25	1.72	1.84	1.99
Y:	50	75.2	42	72.5	90.3	70

6. Obtain / *Verkry*

- Sample size / *Steekproefgrootte* (*n*=6)
- Summations / *Sommasies*

  $\sum_{i=1}^{n} x_i = 9.8$, $\sum_{i=1}^{n} x_i^2 = 16.5466$, $\sum_{i=1}^{n} y_i = 400$, $\sum_{i=1}^{n} y_i^2 = 28229.38$,

  $\sum_{i=1}^{n} x_i y_i = 678.012$

- Averages / *Gemiddeldes*

  $$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i = 1.6333, \quad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i = 66.6667$$

- Standard deviations / *Standaardafwykings*

  $$s_x = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = 0.3286, \quad s_y = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2} = 17.6789$$

- Correlation / *Korrelasie* $r = \frac{s_{xy}}{s_x s_y} = 0.8496$
- *Covariance / Kovariansie* $s_{xy} = r \times s_x \times s_y = 4.9356$